



I&A Report

Customer:	EMISSIONS
Customer PO:	TBD
Part Type:	F7EA 1ST STAGE BUCKETS
APG Job #:	20443
Date:	2026-06-30
Turbine OEM:	GE
Turbine Model:	F7EA
Component:	F7EA 1ST STAGE BUCKETS
Cast Part Number:	109E5281P1H1
Mach Part Number:	109E5289P001
Quantity:	92
Material:	GTD-111

A set of ninety-two (92) F7EA 1ST STAGE BUCKETS was received from EMISSIONS for an Incoming Inspect-and-Advise under APG Job #20443.

Buckets were visually inspected for any distress and incoming-photographed.

Bucket aluminium seal strip and root galling was removed by mechanical methods. Buckets were then dimensional inspected.

Bucket airfoils were cleaned to remove environmental build-up by grit blast, bond coat by chemical stripping methods and heat tinted. Bucket residual coatings were removed by mechanical methods, as required.

Buckets were then FPI/Visual inspected and wall thickness inspected.



FPI & Visual Inspection Summary

FPI & Visual Inspection:

- Performed on all ninety-two (92) buckets per APG procedure documents WI-306 and WI-309.
- Representative photographs were taken of typical indications.

Findings:

- No crack-map source file was found for this job — zone-by-zone finding detail still needs a tech rep's written note before this section is complete.



Dimensional Inspection Summary

TE Angel Wing Height (A):

- Inspected buckets are within APG criteria.

LE Upper Angel Wing Height (B):

- Inspected buckets are within APG criteria.

LE Lower Angel Wing Height (C):

- Inspected buckets are within APG criteria.

Tip Height (D):

- Thirteen (13) of the 20 inspected buckets have at least one reading outside APG criteria.

Squealer Tip Thickness (E):

- Twelve (12) of the 20 inspected buckets have at least one reading outside APG criteria.

Root Serrations Dims:

- Inspected buckets (6 buckets) are within APG criteria. This form samples 6 bucket(s) (APG # 1, 13, 25, 37, 49, 61), not the full population — treat this as a spot-check, not a 100% inspection.

UT Wall Thickness:

- Inspected buckets (92 buckets) are within APG criteria. 0 of 92 bucket(s) have at least one reading below the printed minimum limit.

2 of 7 dimensional categories inspected show at least one reading outside APG criteria: Tip Height (D), Squealer Tip Thickness (E). Overall repairability to be confirmed by the reviewing tech rep.

NOTE: APG will provide best effort to clear all indications within acceptable criteria. There is still the possibility of additional un-repairable components as repairs continue, and further investigation of the distress is determined according to APG.



Recommended Repairs

1. IN625, 0.045" dia
2. Break Platform Edges
3. Prep Indications
4. Air Check Cooling Holes
5. Pre-Weld Buckets
6. Perform a NDT (FPI) Inspection
7. Weld Repair Defects
8. Dust Blast Clean
9. Air Check Cooling Holes
10. Perform a NDT (FPI) Inspection
11. Touch up
12. Foreman Visual Inspection
13. Post Weld Heat Treat
14. Perform a NDT (FPI) Inspection
15. Touch up
16. Thermal Clean
17. Dust Blast Clean
18. Air Check Cooling Holes
19. Local FPI / Visual Inspection
20. Touch up
21. Wall Thickness Inspection
22. Pre Coat Dimensional Inspect
23. Amdry 510 / XPT 510 10#
24. Receive Buckets to Coating
25. Thermal degrease
26. Mask Parts
27. Coat HVOF MCrAlY
28. Clean
29. Open air furnace heat tint
30. Diffusion and Age
31. Air Check Cooling Holes
32. Shot Peen & Al Strip
33. Final Visual Insp & Report
34. Final Dimension
35. Wire Check Cooling Holes
36. Inspect damper pin slots
37. Blend damper pin slots
38. Moment Weigh & Sequence
39. Final Photos
40. Customer Witness



Serial Number List

Customer: EMISSIONS	Date: 3/13/2026	Page: 1
APG Job #: 20443	Insp: Y.M	Cast P/N: 109E5281P1H1
Unit/Frame: F7EA	Row/Stg: 1	Mach P/N: 109E5289P001

APG #	Serial #	Comment	APG #	Serial #	Comment	APG #	Serial #	Comment
1	CIZM149225		32	CIZM151300		63	C1ZM153850	
2	CIZM153796		33	CIZM340881		64	C1ZM156981	
3	CIZM148482		34	CIZP038417		65	C1ZM155187	
4	CIZP038400		35	C1ZP038254		66	C1ZM142530	
5	CIZP038187		36	C1ZP038116		67	C1ZM152463	
6	CIZP038226		37	C1ZP038193		68	C1ZM156523	
7	CIZP038161		38	C1ZP037861		69	C1ZM152900	
8	CIZP038135		39	C1ZP038034		70	C1ZM152099	
9	CIZP038104		40	CIZP038341		71	C1ZM156817	
10	CIZP038339		41	C1ZP038203		72	C1ZM151882	
11	CIZP038321		42	C1ZP038180		73	C1ZM148407	
12	CIZP038325		43	C1ZP038132		74	C1ZM141941	
13	CIZP038376		44	C1ZP038125		75	C1ZM152870	
14	CIZD001310		45	C1ZP038274		76	C1ZM141763	
15	CIZP038234		46	C1ZP038323		77	C1ZM153737	
16	CIZM153001		47	C1ZP038464		78	C1ZM146331	
17	CIZM155985		48	C1ZP038299		79	C1ZM155225	
18	CIZM142123		49	C1ZM153257		80	C1ZM143103	
19	CIZM156248		50	C1ZM153783		81	C1ZM152005	
20	CIZP038351		51	C1ZM153338		82	C1ZM157171	
21	CIZM152080		52	C1ZM155497		83	C1ZM156221	
22	CIZM156450		53	C1ZM153869		84	C1ZM155268	
23	CIZP038155		54	C1ZM156809		85	C1ZM155454	
24	CIZP038130		55	C1ZM141992		86	C1ZM152072	
25	CIZM141151		56	C1ZM153842		87	C1ZM148865	
26	CIZM144436		57	C1ZM153346		88	C1ZM152331	
27	CIZM154814		58	C1ZM156035		89	C1ZM153222	
28	CIZM153729		59	C1ZM150339		90	C1ZM152978	
29	CIZM156795		60	C1ZM146293		91	C1ZM155128	
30	CIZM152110		61	C1ZM155233		92	C1ZM152307	
31	CIZM155489		62	C1ZM146552				

NOTES:



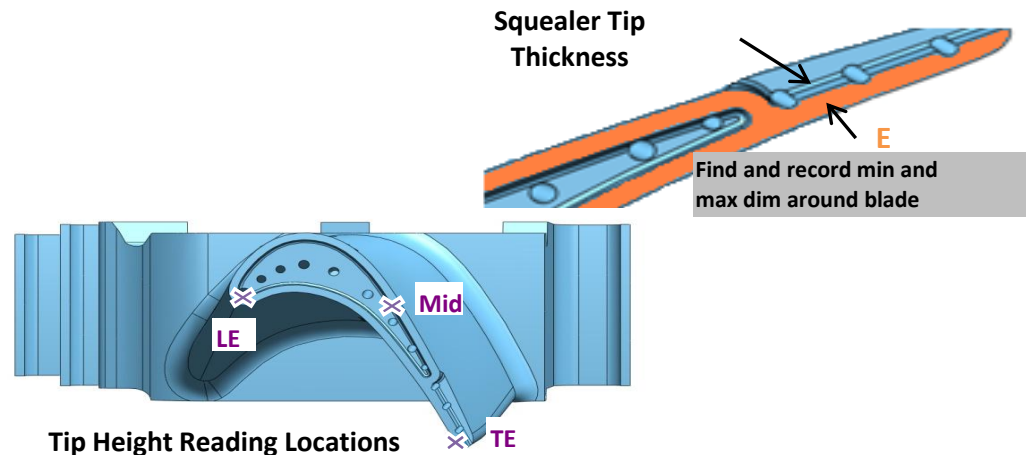
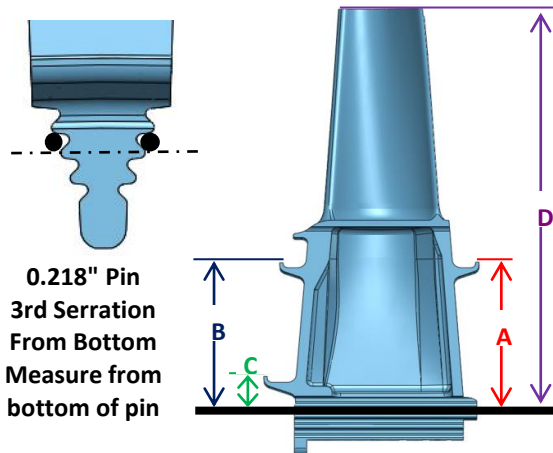
7EA/E/B Row 1 Bucket Height Dim Form

Date: 3/15/2026

Inspector: Y.M

APG Job #: 20443

All Dims +/- 0.025"	A - TE Angel Wing Height	B - LE Upper Angel Wing Height	C - LE Lower Angel Wing Height	D - Tip Height			E - Squealer Tip Thickness	
	4.960" EA 4.960" B	4.960" EA 4.960" B	1.250" EA 1.250" B	13.255" EA 13.226" B	13.250" EA 13.221" B	13.200" EA 13.190" B	0.040" Min	0.125" Max
APG #	MID	MID	MID	LE	MID	TE	Min	Max
1	MM	4.976	1.266	13.251	13.233	13.191	0.031	0.092
2	4.958	4.975	1.256	13.245	13.226	13.181	0.024	0.099
3	4.956	4.970	1.257	13.240	13.212	13.158	0.029	0.088
4	4.958	4.972	1.257	13.260	13.243	MM	0.062	0.078
5	4.959	4.977	1.267	13.255	13.241	13.117	0.070	0.094
6	4.963	4.975	1.269	13.262	13.249	13.143	0.058	0.095
7	4.965	4.974	1.260	13.254	13.246	13.106	0.064	0.099
8	4.960	4.971	1.258	13.258	13.246	13.123	0.026	0.091
9	4.961	4.967	1.251	13.257	13.243	13.129	0.035	0.099
10	4.965	4.978	1.262	13.260	13.247	13.147	0.056	0.097
11	4.957	4.975	1.262	13.261	13.245	13.137	0.039	0.093
12	4.957	4.972	1.262	13.254	13.239	13.178	0.046	0.084
13	4.960	4.977	1.264	13.263	13.251	MM	0.037	0.096
14	4.964	4.983	1.265	13.262	13.236	13.185	0.082	0.097
15	4.957	4.971	1.261	13.260	13.245	13.133	0.048	0.089
16	4.958	4.976	1.267	13.249	13.228	13.181	0.038	0.104
17	4.956	4.974	1.260	13.250	13.225	13.172	0.036	0.085
18	4.956	4.976	1.271	13.247	13.224	13.176	0.038	0.080
19	4.960	4.977	1.264	13.258	13.236	13.174	0.037	0.100
20	4.959	4.966	1.255	13.260	13.245	13.116	0.025	0.091
MIN	4.956	4.966	1.251	13.240	13.212	13.106	0.024	0.078
AVG	4.959	4.974	1.261	13.255	13.238	13.152	0.044	0.092
MAX	4.965	4.983	1.271	13.263	13.251	13.191	0.082	0.104





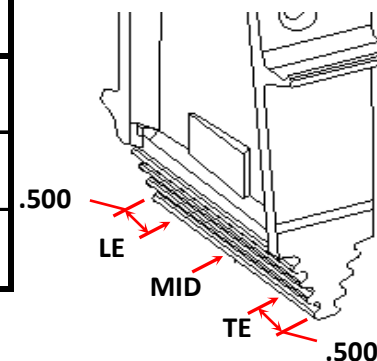
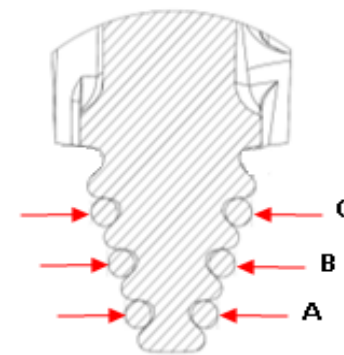
JOB NUMBER 20443

DATE 3/15/26

OPERATOR Y.M

OPERATION INCOMING

APG Bucket #		1	13	25	37	49	61	AVG
A 0.795" 0.815"	LE	0.802	0.798	0.801	0.804	0.803	0.801	0.801
	Mid	0.804	0.804	0.804	0.805	0.807	0.804	0.805
	TE	0.804	0.802	0.804	0.805	0.805	0.803	0.804
B 1.035" 1.055"	LE	1.043	1.041	1.042	1.044	1.043	1.043	1.043
	Mid	1.045	1.044	1.044	1.044	1.045	1.044	1.044
	TE	1.043	1.044	1.042	1.044	1.043	1.041	1.043
C 1.275" 1.295"	LE	1.282	1.279	1.280	1.282	1.282	1.280	1.281
	Mid	1.284	1.284	1.282	1.285	1.282	1.283	1.283
	TE	1.282	1.284	1.281	1.284	1.283	1.281	1.283



TITLE

F7EA Stage 1 Bucket Dovetail Inspection

FORM

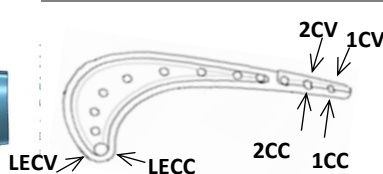
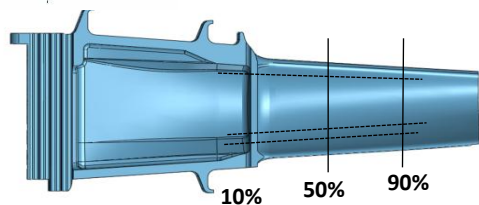
DS-1017

DATE

11/3/2017



7EA Row 1 Bucket Wall Thickness



APG Job#: 20443 Incoming
Date: 5/8/2026
Inspector: IR

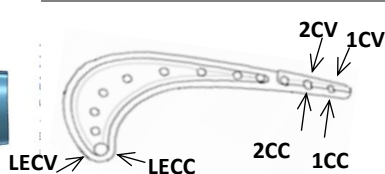
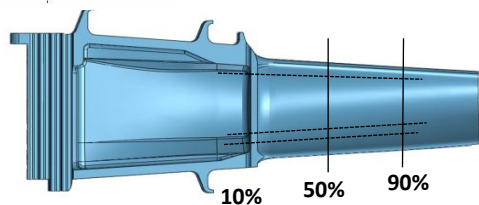
Frame: F7EA

Minimum Limits:

APG #	Height	LECC	LECV	1CC	2CC	1CV	2CV
1	90	0.079	0.090	0.035	0.036	0.036	0.051
	50	0.073	0.102	0.036	0.041	0.065	0.073
	10	0.110	0.102	0.040	0.058	0.070	0.097
2	90	0.075	0.088	0.038	0.039	0.051	0.057
	50	0.066	0.089	0.037	0.051	0.077	0.089
	10	0.102	0.088	0.043	0.065	0.080	0.093
3	90	0.089	0.082	0.046	0.055	0.058	0.062
	50	0.085	0.090	0.045	0.048	0.066	0.076
	10	0.098	0.102	0.038	0.064	0.061	0.083
4	90	0.059	0.093	0.043	0.036	0.048	0.068
	50	0.059	0.083	0.036	0.036	0.060	0.089
	10	0.111	0.083	0.039	0.076	0.065	0.087
5	90	0.064	0.095	0.038	0.040	0.060	0.069
	50	0.064	0.090	0.036	0.039	0.077	0.089
	10	0.093	0.094	0.038	0.071	0.071	0.091
6	90	0.082	0.087	0.039	0.065	0.030	0.036
	50	0.073	0.085	0.036	0.047	0.064	0.071
	10	0.094	0.093	0.055	0.073	0.076	0.084
7	90	0.102	0.085	0.037	0.042	0.038	0.048
	50	0.086	0.099	0.039	0.049	0.046	0.080
	10	0.090	0.107	0.061	0.069	0.060	0.082
8	90	0.109	0.079	0.035	0.047	0.049	0.045
	50	0.099	0.064	0.043	0.054	0.059	0.062
	10	0.117	0.054	0.062	0.089	0.059	0.077
9	90	0.098	0.088	0.038	0.057	0.037	0.054
	50	0.112	0.078	0.039	0.063	0.069	0.081
	10	0.138	0.054	0.062	0.085	0.068	0.087
10	90	0.083	0.084	0.036	0.057	0.045	0.048
	50	0.103	0.083	0.036	0.054	0.077	0.086
	10	0.105	0.088	0.041	0.068	0.077	0.085
11	90	0.077	0.088	0.044	0.045	0.033	0.052
	50	0.086	0.098	0.038	0.061	0.060	0.089
	10	0.094	0.090	0.055	0.084	0.060	0.092
12	90	0.061	0.097	0.038	0.038	0.066	0.072
	50	0.079	0.092	0.037	0.038	0.071	0.083
	10	0.100	0.098	0.050	0.073	0.059	0.085
13	90	0.087	0.083	0.037	0.047	0.031	0.058
	50	0.077	0.088	0.034	0.058	0.059	0.079
	10	0.085	0.079	0.065	0.084	0.056	0.084



7EA Row 1 Bucket Wall Thickness



APG Job#: 20443 Incoming
Date: 5/8/2026
Inspector: IR

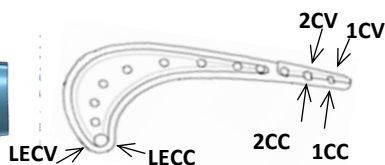
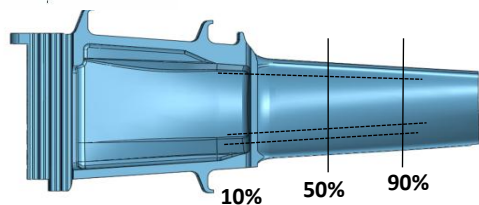
Frame: F7EA

Minimum Limits:

APG #	Height	LECC	LECV	1CC	2CC	1CV	2CV
14	90	0.103	0.095	0.057	0.058	0.057	0.058
	50	0.105	0.095	0.034	0.075	0.093	0.067
	10	0.097	0.106	0.045	0.118	0.090	0.062
15	90	0.104	0.092	0.035	0.035	0.063	0.063
	50	0.111	0.085	0.037	0.041	0.081	0.097
	10	0.111	0.077	0.043	0.070	0.085	0.116
16	90	0.077	0.093	0.043	0.046	0.032	0.034
	50	0.056	0.088	0.036	0.047	0.067	0.076
	10	0.081	0.097	0.048	0.064	0.081	0.090
17	90	0.057	0.095	0.034	0.045	0.055	0.058
	50	0.050	0.095	0.037	0.043	0.077	0.088
	10	0.084	0.095	0.041	0.068	0.075	0.079
18	90	0.070	0.088	0.031	0.043	0.052	0.063
	50	0.062	0.087	0.034	0.035	0.062	0.084
	10	0.091	0.090	0.048	0.050	0.065	0.094
19	90	0.087	0.088	0.039	0.038	0.036	0.035
	50	0.086	0.086	0.039	0.046	0.063	0.068
	10	0.088	0.090	0.048	0.060	0.064	0.082
20	90	0.093	0.092	0.037	0.045	0.052	0.052
	50	0.095	0.087	0.060	0.068	0.056	0.058
	10	0.088	0.087	0.073	0.101	0.036	0.065
21	90	0.057	0.073	0.033	0.039	0.065	0.069
	50	0.075	0.078	0.038	0.037	0.085	0.091
	10	0.093	0.088	0.035	0.052	0.084	0.098
22	90	0.081	0.087	0.036	0.038	0.033	0.035
	50	0.077	0.084	0.038	0.038	0.078	0.082
	10	0.091	0.092	0.039	0.058	0.082	0.091
23	90	0.098	0.095	0.030	0.037	0.057	0.062
	50	0.108	0.089	0.036	0.052	0.071	0.082
	10	0.107	0.091	0.060	0.078	0.069	0.086
24	90	0.079	0.084	0.036	0.058	0.038	0.038
	50	0.079	0.073	0.035	0.036	0.069	0.073
	10	0.090	0.073	0.038	0.062	0.081	0.090
25	90	0.085	0.085	0.036	0.058	0.034	0.036
	50	0.082	0.083	0.036	0.065	0.063	0.066
	10	0.097	0.078	0.038	0.070	0.075	0.079



7EA Row 1 Bucket Wall Thickness



APG Job#: 20443 Incoming
Date: 5/8/2026
Inspector: IR

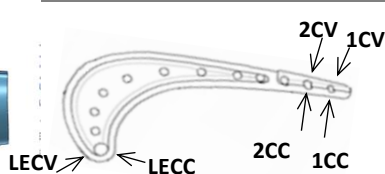
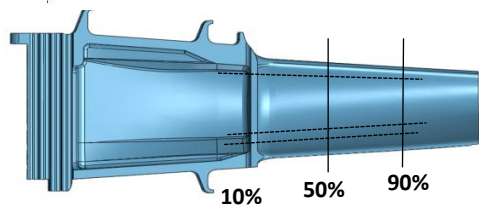
Frame: F7EA

Minimum Limits:

APG #	Height	0.040 LECC	0.040 LECV	0.030 1CC	0.030 2CC	0.030 1CV	0.030 2CV
26	90	0.076	0.078	0.037	0.036	0.044	0.044
	50	0.072	0.087	0.037	0.057	0.069	0.079
	10	0.091	0.096	0.045	0.065	0.068	0.086
27	90	0.064	0.092	0.039	0.035	0.035	0.055
	50	0.054	0.097	0.037	0.036	0.064	0.086
	10	0.080	0.100	0.053	0.062	0.064	0.082
28	90	0.078	0.091	0.036	0.036	0.043	0.046
	50	0.062	0.099	0.042	0.046	0.054	0.074
	10	0.082	0.089	0.038	0.065	0.063	0.089
29	90	0.072	0.095	0.048	0.064	0.046	0.044
	50	0.057	0.099	0.045	0.065	0.070	0.067
	10	0.079	0.090	0.038	0.073	0.078	0.086
30	90	0.078	0.092	0.060	0.062	0.033	0.046
	50	0.075	0.087	0.050	0.071	0.058	0.062
	10	0.096	0.091	0.048	0.067	0.070	0.073
31	90	0.056	0.088	0.035	0.037	0.034	0.046
	50	0.083	0.087	0.033	0.048	0.063	0.070
	10	0.107	0.087	0.038	0.061	0.072	0.080
32	90	0.070	0.081	0.034	0.036	0.054	0.059
	50	0.060	0.091	0.037	0.037	0.076	0.093
	10	0.117	0.091	0.038	0.038	0.087	0.109
33	90	0.090	0.078	0.038	0.045	0.050	0.051
	50	0.094	0.084	0.036	0.044	0.066	0.066
	10	0.108	0.076	0.048	0.066	0.076	0.083
34	90	0.076	0.089	0.035	0.036	0.056	0.056
	50	0.079	0.091	0.037	0.039	0.058	0.084
	10	0.090	0.086	0.052	0.066	0.076	0.088
35	90	0.071	0.086	0.036	0.043	0.048	0.048
	50	0.071	0.086	0.037	0.037	0.063	0.078
	10	0.092	0.098	0.057	0.068	0.078	0.097
36	90	0.076	0.083	0.036	0.035	0.037	0.047
	50	0.079	0.075	0.035	0.048	0.061	0.073
	10	0.105	0.085	0.050	0.074	0.061	0.078
37	90	0.096	0.086	0.034	0.036	0.056	0.057
	50	0.100	0.092	0.037	0.046	0.073	0.083
	10	0.109	0.091	0.040	0.068	0.068	0.095



7EA Row 1 Bucket Wall Thickness



APG Job#: 20443 Incoming
Date: 5/8/2026
Inspector: IR

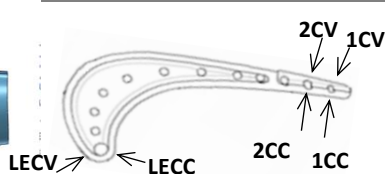
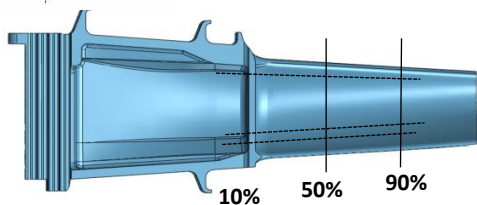
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Minimum Limits:

APG #	Height	LECC	LECV	1CC	2CC	1CV	2CV
38	90	0.082	0.090	0.036	0.038	0.047	0.059
	50	0.075	0.094	0.032	0.061	0.068	0.080
	10	0.093	0.085	0.037	0.096	0.079	0.081
39	90	0.086	0.087	0.036	0.038	0.051	0.049
	50	0.075	0.088	0.037	0.047	0.064	0.071
	10	0.095	0.086	0.038	0.074	0.070	0.088
40	90	0.064	0.073	0.035	0.033	0.037	0.058
	50	0.081	0.072	0.034	0.056	0.066	0.070
	10	0.103	0.069	0.049	0.078	0.059	0.088
41	90	0.083	0.078	0.034	0.047	0.037	0.046
	50	0.085	0.083	0.034	0.046	0.068	0.075
	10	0.089	0.076	0.048	0.071	0.073	0.094
42	90	0.071	0.087	0.032	0.035	0.068	0.069
	50	0.071	0.088	0.034	0.036	0.074	0.089
	10	0.093	0.090	0.058	0.070	0.068	0.078
43	90	0.088	0.089	0.037	0.053	0.045	0.056
	50	0.094	0.089	0.035	0.052	0.070	0.075
	10	0.105	0.081	0.038	0.074	0.063	0.087
44	90	0.072	0.085	0.047	0.045	0.047	0.047
	50	0.074	0.083	0.039	0.054	0.060	0.066
	10	0.097	0.085	0.055	0.076	0.062	0.077
45	90	0.077	0.085	0.032	0.035	0.057	0.058
	50	0.091	0.081	0.039	0.038	0.079	0.085
	10	0.091	0.090	0.061	0.069	0.067	0.097
46	90	0.080	0.083	0.038	0.037	0.045	0.049
	50	0.076	0.089	0.035	0.046	0.066	0.074
	10	0.103	0.088	0.042	0.078	0.086	0.088
47	90	0.059	0.082	0.056	0.066	0.033	0.045
	50	0.062	0.087	0.037	0.060	0.060	0.074
	10	0.086	0.098	0.038	0.084	0.069	0.069
48	90	0.078	0.083	0.034	0.040	0.048	0.055
	50	0.076	0.075	0.034	0.049	0.068	0.079
	10	0.099	0.080	0.037	0.073	0.069	0.092
49	90	0.073	0.093	0.040	0.047	0.049	0.046
	50	0.086	0.086	0.046	0.073	0.057	0.089
	10	0.098	0.091	0.045	0.087	0.077	0.088



7EA Row 1 Bucket Wall Thickness



APG Job#: 20443 Incoming
Date: 5/8/2026
Inspector: IR

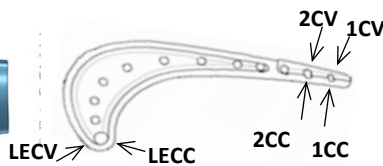
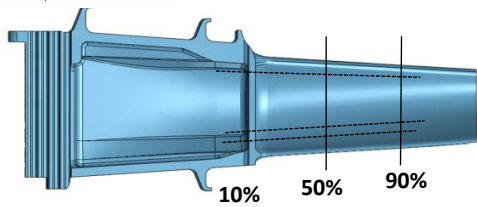
Frame: F7EA

Minimum Limits:

APG #	Height	LECC	LECV	1CC	2CC	1CV	2CV
50	90	0.065	0.092	0.047	0.043	0.058	0.056
	50	0.063	0.088	0.041	0.079	0.054	0.090
	10	0.088	0.100	0.041	0.084	0.045	0.094
51	90	0.058	0.094	0.068	0.046	0.065	0.054
	50	0.080	0.092	0.050	0.083	0.063	0.086
	10	0.094	0.098	0.049	0.078	0.072	0.089
52	90	0.081	0.092	0.041	0.062	0.048	0.065
	50	0.082	0.094	0.048	0.088	0.061	0.084
	10	0.086	0.098	0.046	0.088	0.077	0.081
53	90	0.084	0.093	0.054	0.043	0.061	0.054
	50	0.073	0.089	0.045	0.076	0.058	0.088
	10	0.091	0.087	0.045	0.086	0.066	0.104
54	90	0.082	0.092	0.042	0.042	0.053	0.045
	50	0.083	0.093	0.046	0.069	0.063	0.070
	10	0.082	0.090	0.053	0.066	0.085	0.076
55	90	0.081	0.091	0.041	0.045	0.055	0.055
	50	0.072	0.103	0.043	0.074	0.052	0.086
	10	0.083	0.098	0.046	0.077	0.054	0.099
56	90	0.078	0.098	0.038	0.047	0.050	0.056
	50	0.064	0.094	0.043	0.062	0.056	0.083
	10	0.089	0.083	0.046	0.086	0.061	0.091
57	90	0.083	0.090	0.043	0.045	0.053	0.052
	50	0.091	0.088	0.044	0.064	0.053	0.072
	10	0.100	0.084	0.047	0.080	0.075	0.097
58	90	0.081	0.088	0.039	0.056	0.044	0.059
	50	0.079	0.087	0.042	0.075	0.056	0.077
	10	0.093	0.079	0.042	0.077	0.071	0.079
59	90	0.089	0.088	0.034	0.047	0.037	0.045
	50	0.085	0.098	0.037	0.073	0.068	0.074
	10	0.098	0.090	0.040	0.087	0.073	0.069
60	90	0.078	0.078	0.036	0.035	0.033	0.052
	50	0.076	0.083	0.037	0.038	0.058	0.058
	10	0.099	0.076	0.057	0.069	0.070	0.065
61	90	0.078	0.084	0.036	0.035	0.043	0.069
	50	0.062	0.083	0.035	0.036	0.054	0.089
	10	0.082	0.088	0.050	0.062	0.063	0.078



7EA Row 1 Bucket Wall Thickness



APG Job#: 20443 Incoming
 Date: 5/8/2026
 Inspector: IR

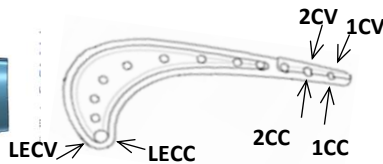
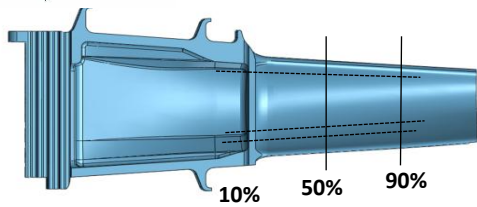
Frame: F7EA

Minimum Limits:

APG #	Height	0.040 LECC	0.040 LECV	0.030 1CC	0.030 2CC	0.030 1CV	0.030 2CV
62	90	0.078	0.089	0.056	0.036	0.050	0.055
	50	0.076	0.091	0.037	0.057	0.066	0.086
	10	0.099	0.086	0.038	0.065	0.076	0.082
63	90	0.082	0.095	0.034	0.043	0.035	0.057
	50	0.075	0.099	0.034	0.035	0.064	0.083
	10	0.093	0.090	0.048	0.050	0.064	0.095
64	90	0.090	0.092	0.056	0.036	0.033	0.047
	50	0.094	0.097	0.037	0.046	0.058	0.066
	10	0.108	0.100	0.038	0.065	0.070	0.077
65	90	0.070	0.088	0.038	0.040	0.037	0.055
	50	0.062	0.098	0.035	0.049	0.068	0.071
	10	0.091	0.090	0.042	0.073	0.073	0.088
66	90	0.086	0.088	0.031	0.035	0.033	0.059
	50	0.075	0.098	0.034	0.036	0.058	0.093
	10	0.095	0.090	0.048	0.070	0.070	0.109
67	90	0.071	0.088	0.035	0.047	0.068	0.045
	50	0.071	0.087	0.033	0.054	0.074	0.062
	10	0.093	0.090	0.038	0.089	0.068	0.077
68	90	0.086	0.082	0.032	0.033	0.063	0.062
	50	0.075	0.087	0.039	0.056	0.081	0.082
	10	0.095	0.098	0.061	0.078	0.085	0.086
69	90	0.086	0.095	0.038	0.047	0.031	0.038
	50	0.075	0.095	0.039	0.046	0.059	0.073
	10	0.095	0.106	0.062	0.071	0.056	0.090
70	90	0.109	0.081	0.036	0.047	0.048	0.047
	50	0.099	0.091	0.036	0.073	0.060	0.073
	10	0.117	0.091	0.038	0.087	0.065	0.078
71	90	0.070	0.086	0.031	0.036	0.037	0.034
	50	0.060	0.092	0.034	0.036	0.066	0.076
	10	0.117	0.091	0.048	0.076	0.059	0.090
72	90	0.077	0.083	0.035	0.036	0.050	0.035
	50	0.056	0.089	0.037	0.037	0.066	0.068
	10	0.081	0.088	0.052	0.038	0.076	0.082
73	90	0.066	0.098	0.038	0.061	0.054	0.057
	50	0.078	0.109	0.053	0.080	0.073	0.085
	10	0.123	0.127	0.086	0.095	0.097	0.108



7EA Row 1 Bucket Wall Thickness



APG Job#: 20443 Incoming
Date: 5/8/2026
Inspector: IR

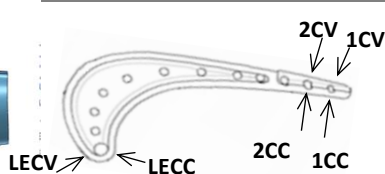
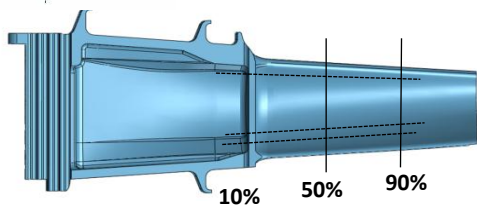
Frame: F7EA

Minimum Limits:

APG #	Height	LECC	LECV	1CC	2CC	1CV	2CV
74	90	0.065	0.084	0.044	0.059	0.052	0.055
	50	0.085	0.098	0.054	0.065	0.078	0.088
	10	0.087	0.106	0.055	0.069	0.096	0.112
75	90	0.068	0.057	0.033	0.042	0.042	0.044
	50	0.062	0.084	0.052	0.069	0.069	0.083
	10	0.091	0.099	0.074	0.102	0.081	0.092
76	90	0.076	0.082	0.050	0.052	0.058	0.059
	50	0.070	0.086	0.069	0.082	0.081	0.085
	10	0.103	0.092	0.072	0.096	0.095	0.106
77	90	0.077	0.082	0.035	0.054	0.056	0.060
	50	0.081	0.097	0.050	0.067	0.073	0.090
	10	0.085	0.103	0.069	0.089	0.074	0.097
78	90	0.080	0.088	0.045	0.060	0.037	0.038
	50	0.088	0.097	0.046	0.069	0.082	0.092
	10	0.096	0.102	0.071	0.076	0.093	0.102
79	90	0.080	0.081	0.036	0.040	0.058	0.063
	50	0.052	0.077	0.045	0.052	0.085	0.090
	10	0.092	0.107	0.069	0.056	0.085	0.107
80	90	0.088	0.096	0.045	0.046	0.048	0.055
	50	0.078	0.099	0.068	0.089	0.079	0.089
	10	0.095	0.123	0.079	0.093	0.090	0.107
81	90	0.074	0.091	0.046	0.056	0.054	0.074
	50	0.075	0.097	0.047	0.058	0.087	0.090
	10	0.095	0.106	0.050	0.066	0.081	0.103
82	90	0.070	0.093	0.033	0.054	0.044	0.051
	50	0.078	0.090	0.042	0.056	0.052	0.061
	10	0.082	0.098	0.047	0.068	0.078	0.105
83	90	0.080	0.093	0.054	0.065	0.044	0.045
	50	0.064	0.073	0.038	0.058	0.054	0.062
	10	0.097	0.105	0.058	0.072	0.078	0.109
84	90	0.070	0.082	0.042	0.052	0.074	0.076
	50	0.064	0.076	0.051	0.057	0.079	0.092
	10	0.087	0.084	0.059	0.062	0.096	0.106
85	90	0.077	0.092	0.031	0.033	0.036	0.052
	50	0.091	0.087	0.040	0.057	0.081	0.088
	10	0.091	0.091	0.054	0.078	0.095	0.112



7EA Row 1 Bucket Wall Thickness

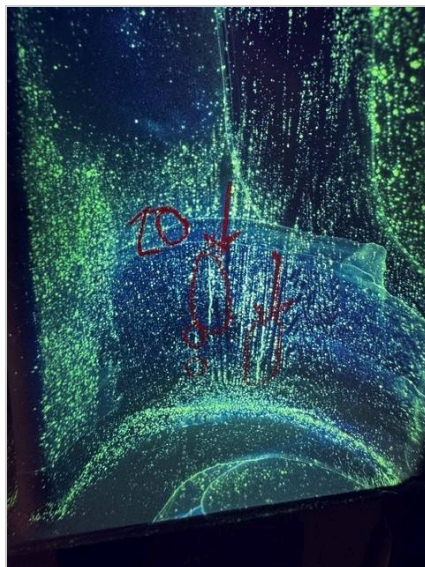
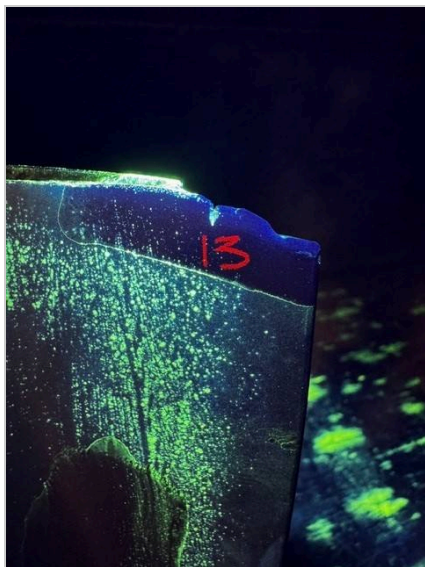
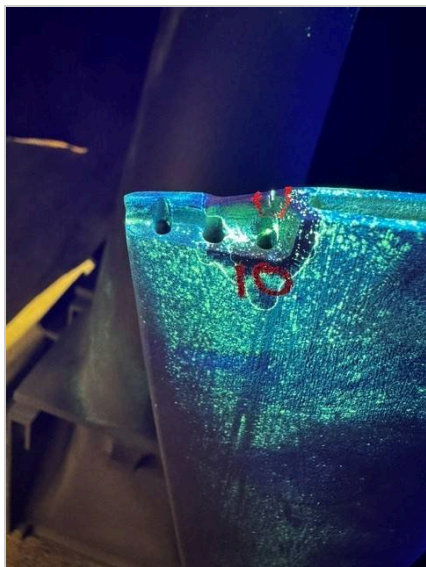
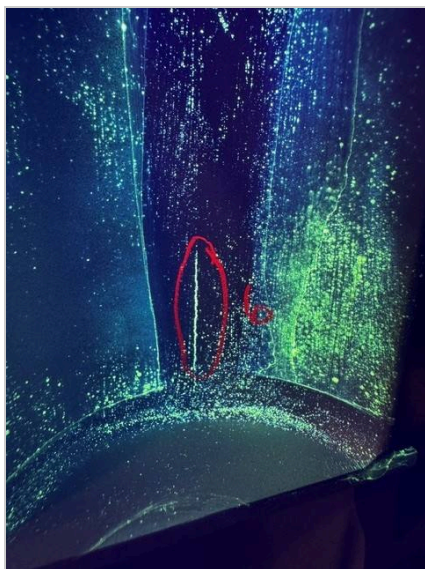
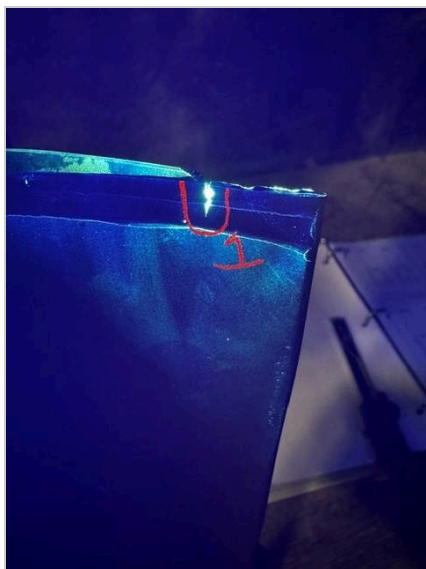


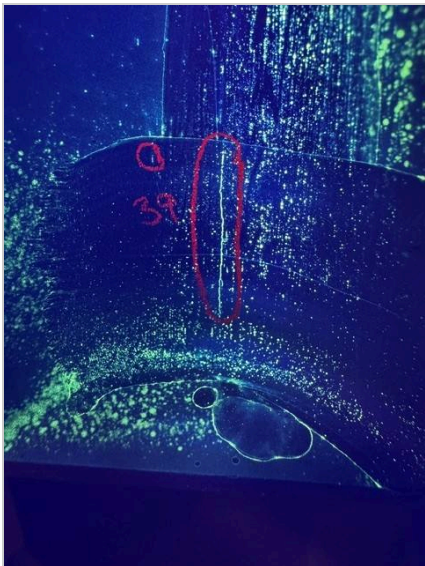
APG Job#: 20443 Incoming
 Date: 5/8/2026
 Inspector: IR

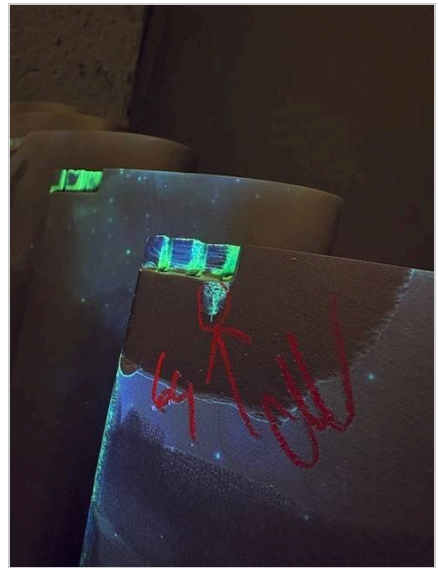
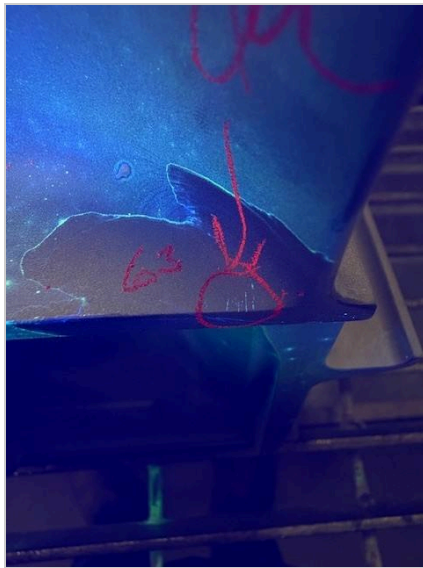
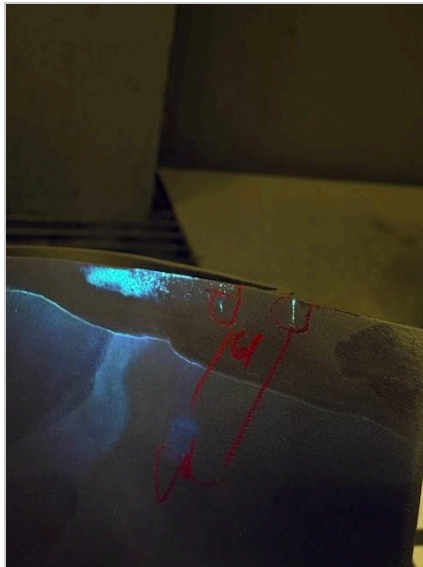
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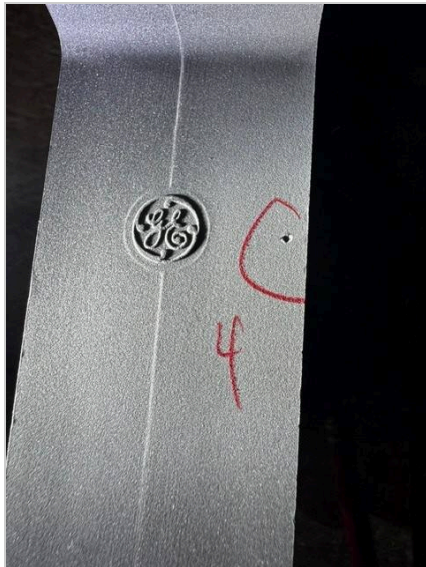
Minimum Limits:

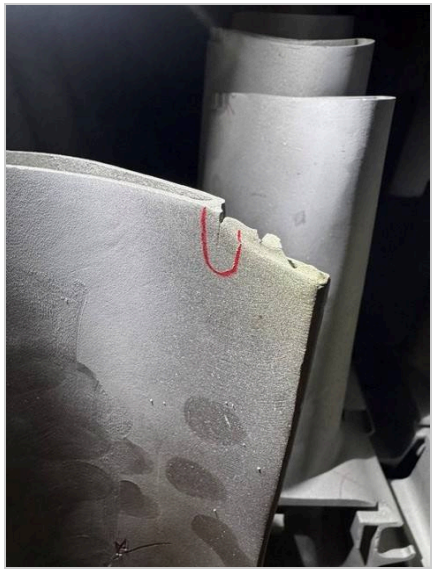
APG #	Height	LECC	LECV	1CC	2CC	1CV	2CV
86	90	0.082	0.056	0.042	0.044	0.067	0.068
	50	0.072	0.089	0.048	0.052	0.086	0.089
	10	0.096	0.081	0.052	0.069	0.096	0.106
87	90	0.046	0.057	0.042	0.052	0.059	0.079
	50	0.064	0.093	0.048	0.055	0.078	0.082
	10	0.098	0.096	0.047	0.068	0.084	0.096
88	90	0.066	0.082	0.035	0.052	0.062	0.065
	50	0.073	0.096	0.041	0.060	0.082	0.089
	10	0.089	0.100	0.045	0.077	0.094	0.105
89	90	0.072	0.094	0.048	0.050	0.062	0.065
	50	0.053	0.081	0.049	0.052	0.081	0.096
	10	0.096	0.092	0.065	0.078	0.097	0.120
90	90	0.073	0.089	0.039	0.042	0.068	0.073
	50	0.083	0.092	0.046	0.053	0.086	0.096
	10	0.102	0.102	0.056	0.063	0.098	0.104
91	90	0.071	0.091	0.036	0.045	0.032	0.046
	50	0.067	0.092	0.048	0.063	0.071	0.075
	10	0.084	0.095	0.050	0.056	0.087	0.098
92	90	0.078	0.042	0.034	0.052	0.037	0.047
	50	0.081	0.096	0.051	0.056	0.078	0.088
	10	0.092	0.113	0.059	0.063	0.095	0.108
MIN		0.046	0.042	0.030	0.033	0.030	0.034

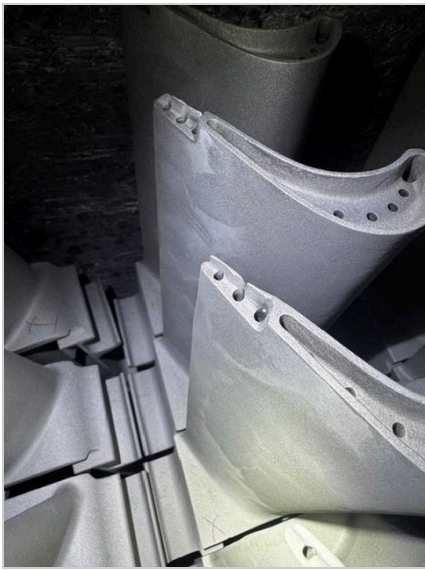
















Crack Map

No crack map was found for this job.



Gas Turbine Metallurgical Report 2026-139

Metallurgical Evaluation of a 7EA 1st-Stage Bucket (GTD-111 DS)

Client / PO #: Allied Power Group / 0062046 - 00

Equipment Owner: Emissions

Job #	Bucket #	Serial Number
20443	1	C1ZM-149225

Date: March 22, 2026

INTRODUCTION

Allied Power Group sent one sample cut from a GE 7EA 1st-stage bucket to VeriTech Labs for evaluation of the microstructure and coating in the service-run condition. The sample was cut from the leading edge (LE) tip from bucket position #1, SN C1ZM-149225, which is made from DS alloy GTD-111. This bucket has no previous repairs. The fired hours and starts were not provided.

The sample was mounted in epoxy and prepared for optical microscopy by grinding, polishing and etching in a nitric/lactic/HF acid mixture.

MICROSTRUCTURES

The LE tip gas-path surface in Figure 1a displays a ceramic top coat 13–19 mils thick and an HVOF MCrAlY bond coat 4–8 mils thick. The base metal has been well protected.

The uncoated LE cooling-hole surface in Figure 1b shows severe depletion at the surface 0.5–1.5 mils deep and moderate depletion 1–2 mils deep below the surface layer. The alloy migration layer is 1.5–3 mils thick. The base metal is unaffected under the alloy migration layer.

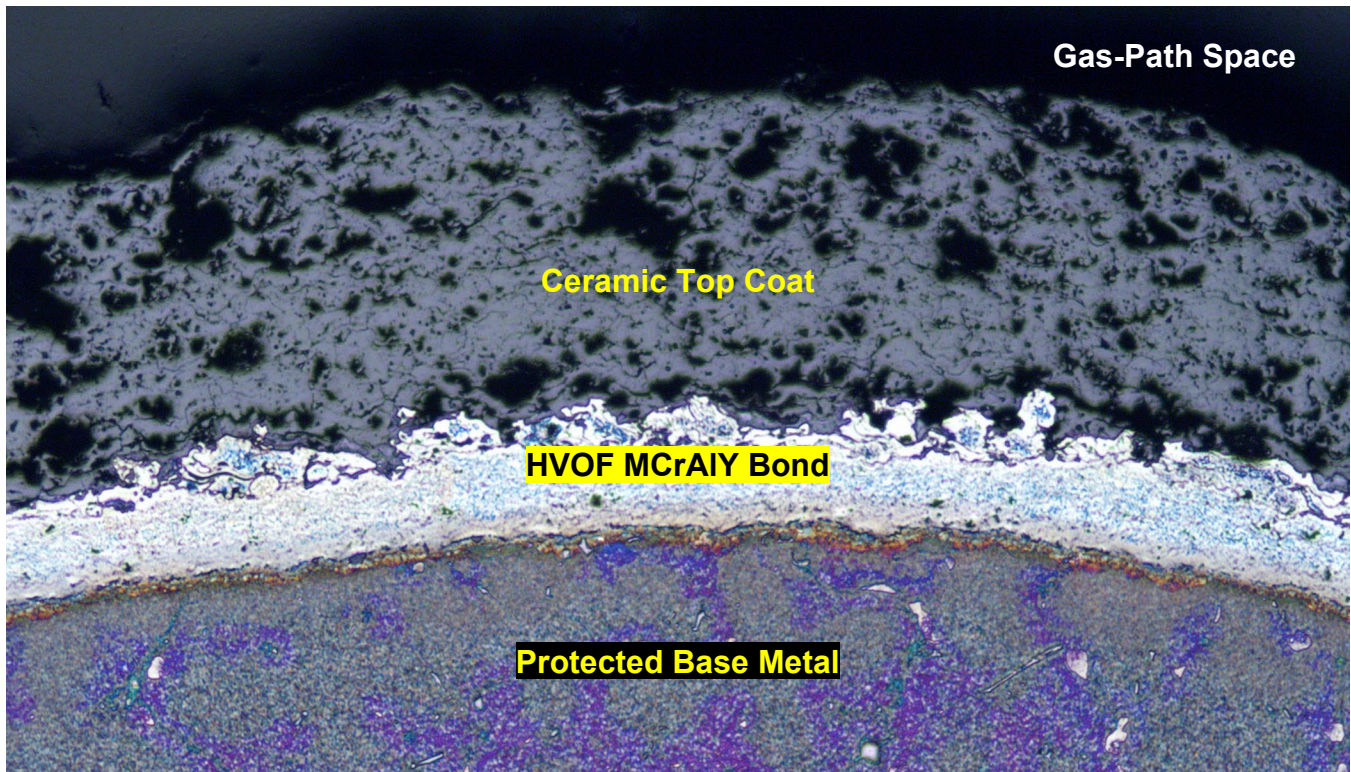
The bulk microstructures in Figure 2 show the following features:

- The fine primary gamma prime (γ') particles in the matrix are not aged.
- There are numerous primary γ - γ' eutectic islands throughout the matrix.
- There are large primary γ' particles in the matrix.
- The MC carbides are stable.

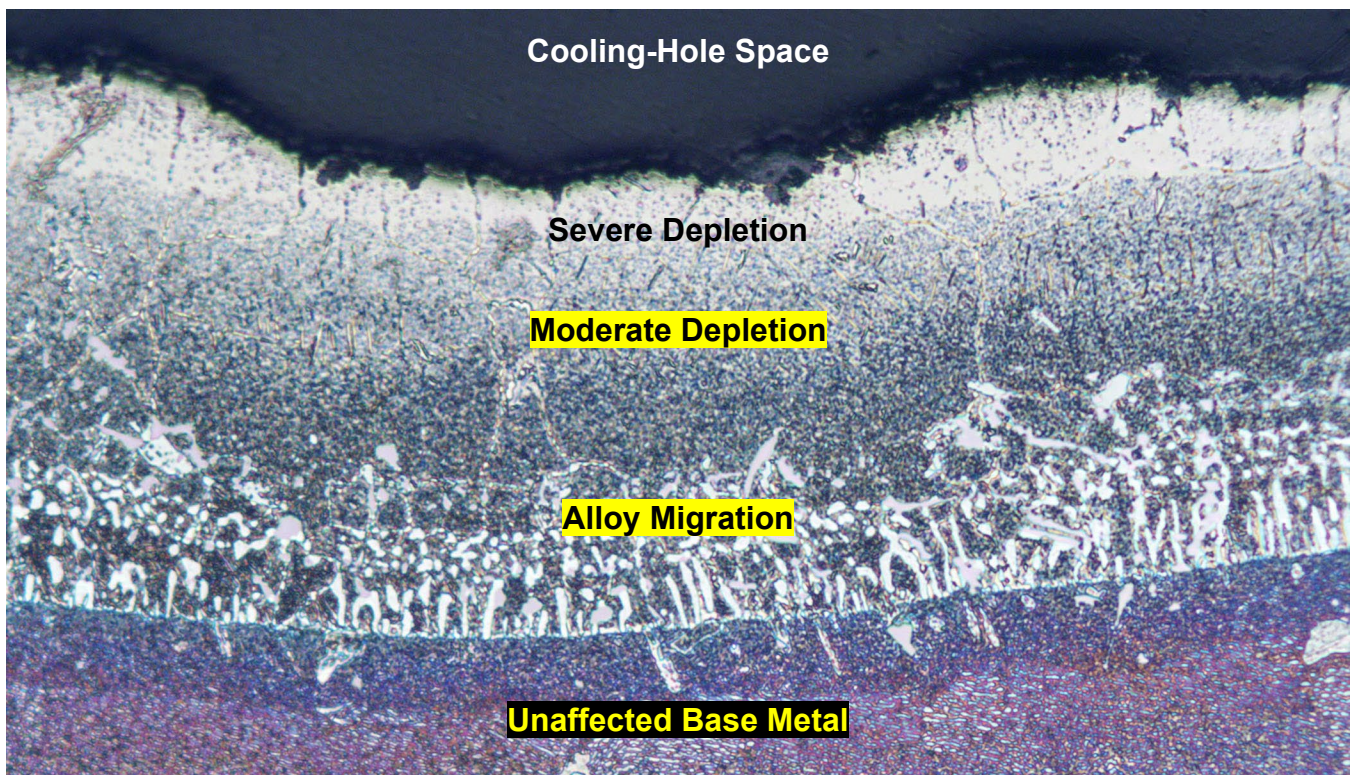
CONCLUSIONS

1. The gas-path has been well protected by the coating.
2. The depletion in the cooling hole is severe at the surface, but the depth is not severe.
3. The primary γ' is not aged. The MC carbides have not started to decompose.
4. Based on the submitted sample, this bucket set is expected to have excellent serviceability after the repairs are complete.



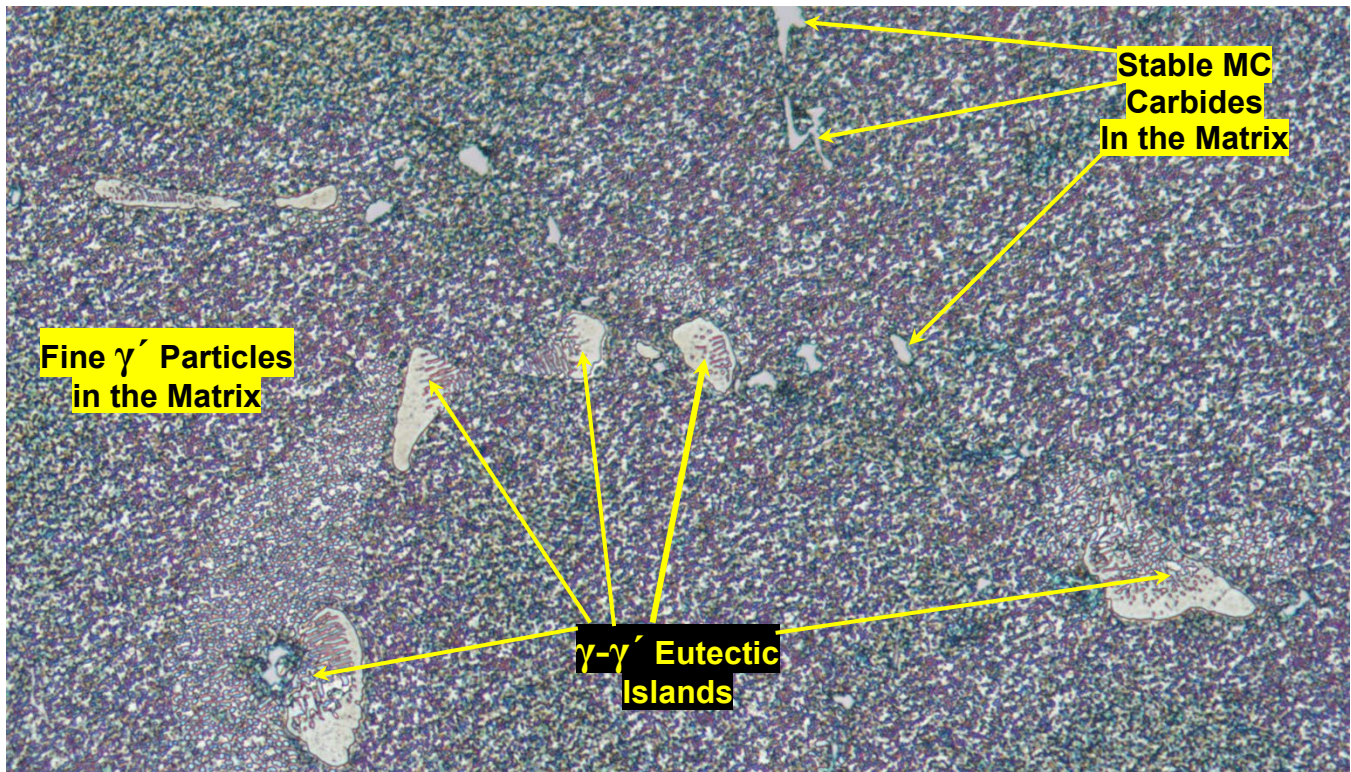


a) Gas-Path Side, 100x

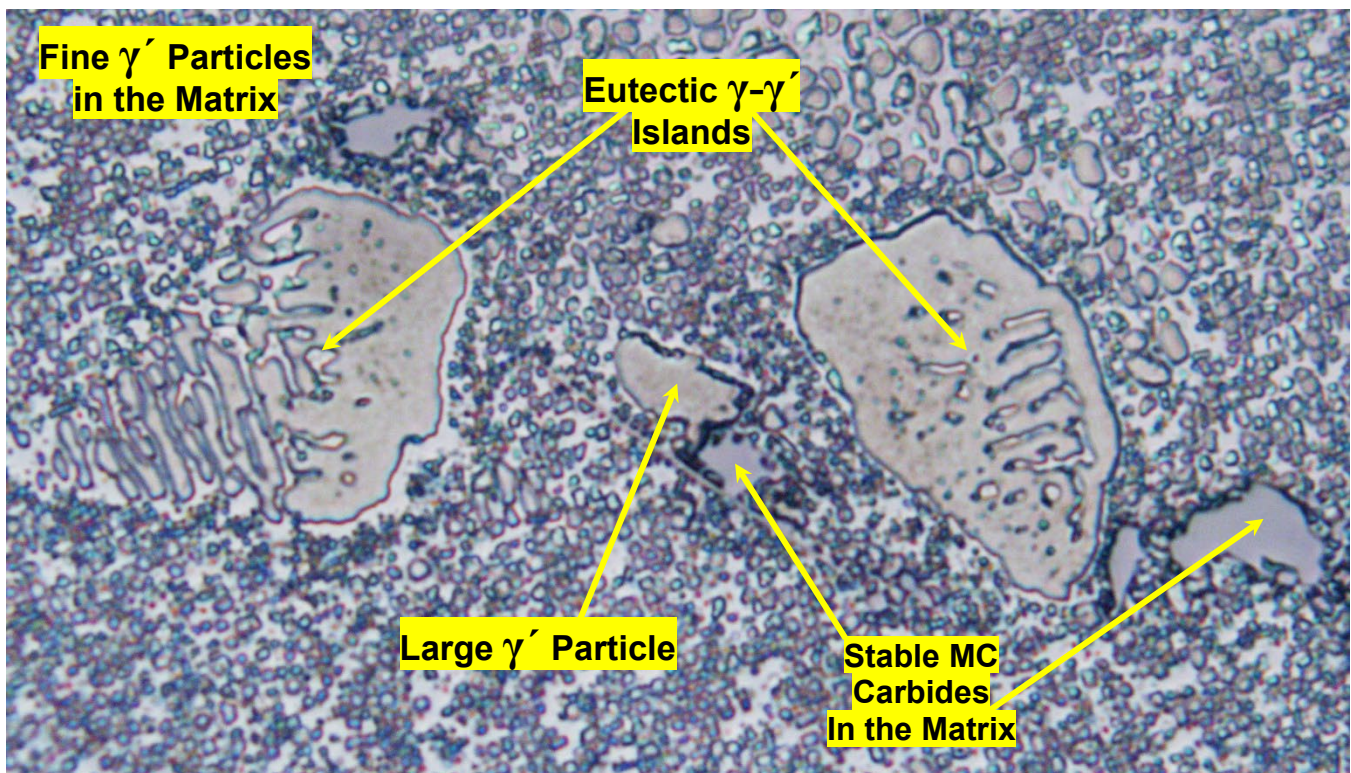


b) Cooling-Hole Side, 400x

Figure 1: Bucket 1 Service-Run Surface Microstructures; Nitric/Lactic/HF Etch.



a) General Microstructure, 400x



b) Detailed Microstructure, 2,000x

Figure 2: Bucket 1 Service-Run Bulk Microstructures. Nitric/Lactic/HF acid Etch.

Gas Turbine and Steam Turbine Inspections

Stress-Rupture and Creep-Rupture Testing

Gas Turbine Hot Section Life Assessment

In-Situ Metallurgy and Hardness Testing

Welding Procedures and Qualifications

Microscopic Thickness Measurements

Metallurgical and Coating Evaluation

Hardness and Microhardness Testing



Flame Spray North America Inc

576 International Blvd.
Fountain Inn, SC 29644
United States
Phone: (864) 862-2730
Fax: (864) 862-7465

Certificate Of Conformance

To: ALLIED POWER GROUP
10131 Mills Road
Houston, TX 77070
United States

From: Flame Spray North America Inc
576 International Blvd.
Fountain Inn, SC 29644
United States

Packing List Number: 8879
Shipping Date: 04/08/26

Today's Date: 04/09/26
PO Number: 0062085 - 00 Job 20443

Quantity

<u>Shipped</u>	<u>Unit</u>	<u>Description</u>	<u>Revision</u>	<u>Job Number</u>
92	EA	F7ES1B 7E Stage 1 Bucket Chem Strip	00	3371-01

A handwritten signature in black ink, appearing to read "Robert Ravelo", written over a light blue horizontal line.

Robert Ravelo

Inspector's Initials
Flame Spray North America Inc